

Francesco Zambelli

PhD student in Physics, University of Padova

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Verona, Italy

+39 3462208470

francescozambelli

francesco.zambelli98@gmail.com

francesco-zambelli-98it

About me

I am a 26-year-old third-year PhD candidate working in the field of complexity sciences, with a focus on biological networks within the broader framework of complex networks. I am part of the CoMuNe Lab, where I contribute to research on the structure, dynamics, and function of complex systems. Beyond my academic interests, I am enthusiastic about continuous learning and enjoy creative and social activities such as singing, dancing, and connecting with people.

Education

2023 - CURR. Ph.D. in Physics — University of Padova (IT)

I am a member of the CoMuNe Lab, led by Prof. Manlio De Domenico, which studies complex networks and complex systems across a wide range of applications. My research focuses on biological networks, particularly gene regulatory networks and the inference methods used to construct networks from experimental data. I am also interested in modeling intracellular biological processes and evolutionary dynamics. I spent a 6 month period in the Complexity Systems lab of Prof. Ricard Solé working on cancer evolutionary dynamics and statistical mechanics formulation of liquid brain systems.

2020-2023 M.Sc. in Physics of Data, 110/110 cum Laude — University of Padova (IT)

This program provides theoretical and practical training to connect physics research with real-world applications. My focus combined statistical mechanics and complex systems with courses on machine learning and AI, emphasizing data analysis of complex and heterogeneous datasets.

2017-2020 B.Sc. in Fisica, 107/110 — University of Padova (IT)

It offers students a broad and comprehensive overview of the fundamental topics in physics and mathematics.

2011 - 2017 High school, 100/100 — Liceo Scientifico N. Copernico (VR)

Publications

2025 Self-Organized cancer fragility emerges from chromosomal instability

In progress

2025 Unraveling the network signatures of oncogenicity in virus-human protein-protein interactions — Submitted on *Entropy, MDPI*

2025 Challenges and opportunities for digital twins in precision medicine from a complex systems perspective

2022 Natural climate reconstruction in the Norwegian stave churches through time series processing with variational autoencoders

Conferences and Seminars

2024 MSCX 2024, Mediterranean School of Complex Networks, Grado (IT)

Regular talk about *"Unraveling the network signatures of oncogenicity in virus-human protein-protein interactions"*

2023 NetSciX 2024, International School and Conference on Network Science, Venice (IT)

Regular talk about *"Characterizing viral oncogenic features from protein-protein multilayer interaction networks"*

2022 Spring Workshop on Physics of Data, Venice (IT)

Workshop focused on presenting how the application of Machine Learning techniques and Artificial Intelligence are contributing to the development of scientific research in many fields, such as cosmology, neuroscience, particle physics

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Driving Licence

Italian driving licence of type B

Digital skill

Python, R Git, GitHub, Linux environment user, Machine Learning and Deep Learning in Python, Latex, Using remote-cloud computational resources (Google Cloud Platform, ssh protocols)

Languages skill

Spanish



Italian



English



Programming skill

Python



R



Matematica



Experiences

2025 — **Abroad research period** Parc de Recerca Biomèdica de Barcelona (UPF), Barcelona, SP

I spent six months in Barcelona collaborating with Prof. Ricard Solé, head of the Complex Systems Lab at UPF. During this period, I worked on developing mechanistic models to simulate and analyze tumor tissue expansion, as well as studying the mathematical formulation of the computational abilities of liquid systems.

since 2023 — **PhD student in CoMuNe Lab** University of Padova

I am currently a PhD researcher, developing the ability to independently organize my work with guidance from my supervisor and lab members, while staying productive and up to date with relevant research developments.

2023 — **Internship abroad** CRCT (Inserm), Tolouse, FR

I spent three months at an oncology research center working on my Master's thesis, exploring biological topics in depth and applying my technical skills to support ongoing bioinformatics research projects within the host team.

since 2019 — **Family's rents manager**

For the past years, I have managed most aspects of renting several family properties, including advertising, meeting prospective tenants, monitoring payments, and resolving issues by implementing appropriate solutions.

since 2018 — **Volunteering**

I have spent several years as a parish educator for teenagers, coordinating a team of educators to organize weekly meetings, charity events, and trips for 40–70 participants. This role requires strong communication skills across different age groups to foster intergenerational connections. It has been a highly formative experience, enhancing my leadership, creativity, and problem-solving abilities.